

PAUL NIGEL S. ABALOS, DIT

Full-Stack Developer | IT Educator | Researcher

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PROFESSIONAL SUMMARY

Full-Stack Developer and Information Technology Educator with nearly a decade of higher-education experience and hands-on experience building practical web applications, real-time PWAs, transactional database systems, and community-oriented digital solutions. Strong in PHP, MySQL, JavaScript, REST APIs, responsive web development, database design, payment/API integrations, and automated testing. Also experienced in academic research, data analytics, bibliometrics, AI/ML applications, and research writing. Combines software engineering, teaching, and research to turn real-world problems into usable technology solutions.

CORE TECHNICAL SKILLS

Backend & Web: PHP 8+, PDO, Laravel, JavaScript ES6+, Vanilla JS, AJAX, HTML5, CSS3, Bootstrap 5, Tailwind CSS, jQuery, Blade, Responsive Web Design

Database & Systems: MySQL/MariaDB, InnoDB, relational database design, ERD, normalization, SQL, concurrency control, transactions, RESTful APIs, webhooks

Integrations & Testing: PayMongo API, Revolut Pay, OCR.space, Brevo API, Leaflet/OpenStreetMap, Playwright, Axe-Core, PWA, Service Workers, Web Speech API, jsQR

Networking & Tools: Cisco Networking, MikroTik RouterOS, LAN/WAN, IP addressing/subnetting, Git, GitHub, Composer, VS Code, XAMPP, phpMyAdmin

Research & AI: Research methodology, academic writing, quantitative research, AI/ML, data analytics, bibliometric analysis, SPSS, Bibliometrix, VOSviewer, OpenAlex, Scopus, Google Scholar

PROFESSIONAL EXPERIENCE

Assistant Professor I | Pangasinan State University – San Carlos Campus | September 2024 – Present

- Teach undergraduate IT courses in web development, web systems and technologies, operating systems, computer networks, database management, discrete mathematics, and introductory Laravel development.
- Design instructional modules, laboratory manuals, and technology-enhanced learning resources; mentor capstone teams through analysis, design, development, testing, and documentation.
- Develop and maintain web-based applications for academic and institutional initiatives while conducting scholarly research and contributing to peer-reviewed publications.

Instructor I | Pangasinan State University – San Carlos Campus | August 2016 – September 2024

- Delivered project-based and laboratory-centered instruction in software development, database design, networking, and systems analysis.
- Supervised capstone projects and undergraduate research and produced instructional materials supporting curriculum delivery and student learning.
- Contributed to academic and departmental initiatives promoting technology-enhanced education.

SELECTED DEVELOPMENT PROJECTS

PickleDink – Real-Time Sports Community PWA | Full-Stack Developer

- Built a production-ready court queue and analytics system for a community of competitive pickleball players, providing a free alternative to subscription-based tools.
- Implemented automated court rotation, dynamic Elo rating/leaderboards, TV announcer mode using Web Speech API, QR-based check-in, and targeted silent push notifications.
- Stack: PHP, MySQL, Vanilla JavaScript, AJAX, Bootstrap, PWA/Service Workers, Web Speech API, jsQR.

SakayPH – Carpool & Chauffeur Platform | Full-Stack Developer

- Designed a database-driven transportation platform with secure escrow, document verification, driver/passenger workflows, and payment integration.
- Implemented 24-hour automated escrow release, InnoDB row-level locking with SELECT ... FOR UPDATE, immutable double-entry transaction ledger, and OCR-based license/OR-CR verification.
- Stack: PHP/PDO, MySQL InnoDB, PayMongo API, OCR.space API, Brevo API, Leaflet/OpenStreetMap.

La Mesa – Restaurant & Delivery Logistics Platform | Full-Stack Developer

- Developed a production web platform for a restaurant business in Guernsey, UK, with separate Super Admin, Owner, Rider, and Customer portals.
- Implemented postcode/parish validation, AJAX ordering workflows, delivery/pickup fee calculations, payment confirmation tools, QR payments, and automated order tracking.
- Added Playwright end-to-end testing and Axe-Core accessibility audits for checkout and ordering workflows.

PSU Pageant Tabulation System | System Analyst / Co-Developer

- Conducted requirements analysis and contributed to development, testing, and deployment of a role-based web tabulation system with candidate management, weighted scoring, real-time results, and report generation; covered by an IPOPHL copyright registration.

RESEARCH & PUBLICATIONS

- Research interests include artificial intelligence, information systems, educational technology, data analytics, bibliometric analysis, recommendation models, and digital learning transformation.
- AI-Based Resume Skills Extractor and Recommender Modules for State University Human Resource Analytics System (2024).
- A Data-Driven Profile-Based Analytics for Career Path and Upskilling Recommendations in HRS (2024).
- The Influence of Artificial Intelligence on Human Decision-Making, Productivity, and Safety in the Field of Education at Pangasinan State University (2026).
- Assessment of Online Learning System to College Students of State Universities in Region I (2023).
- Microsoft Teams Utilization for Teaching and Learning During Pandemic in Pangasinan State University.

EDUCATION

- Doctor of Information Technology (DIT) | University of the Cordilleras
- Master of Information Technology | Colegio de Dagupan | 2016–2019
- Bachelor of Science in Information Technology | Colegio de Dagupan | 2012–2016

PROFESSIONAL POSITIONING

Open to: Part-time, freelance, remote, and project-based opportunities in full-stack web development, information systems, technology projects, research/analytics, and educational technology.